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A Soft Iterative Receiver With Simplified EP Detection for Coded MIMO Systems

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Abstract—Expectation propagation (EP) achieves excellent performance with high-order modulation in massive multiple-input multiple-output (MIMO) detection. The soft output of the EP detector can be iteratively combined with turbo soft decoders to enhance error-correction performance. However, the implementation of EP-based iterative detection and decoding (IDD) receivers suffer from an exponential increase in computational complexity as the number of antennas and modulation order grows. In this brief, we propose a simplified EP approximation-based IDD (sEPA-IDD) scheme for hardware implementation. To alleviate the computational burden, a simplified message update scheme is proposed, reducing complexity by 68% without performance degradation. Additionally, a unified design for extrinsic message computation further improves hardware utilization. Finally, we introduce the first unfolded EP-based IDD architecture to boost throughput. Compared with state-of-the-art (SOA) IDD receivers, the sEPA-IDD receiver implemented on 65 nm CMOS delivers a throughput of 3.07 Gb/s with a maximum 0.5 dB gain, achieving 4.03× higher throughput and 6.04× greater area efficiency.

Index Terms—Expectation propagation (EP), hardware implementation, iterative detection decoding, massive multiple-input multiple-output (MIMO), polar codes.

I. INTRODUCTION

Massive multiple-input multiple-output (MIMO) offers high spectrum utilization efficiency and data rates for wireless communication. To further improve error-correction performance, coded MIMO systems have attracted significant attention, leading to the exploration of efficient iterative detection and decoding (IDD) schemes. However, high hardware cost due to the increasing number of antennas and modulation orders impedes the practical application of IDD receivers. Hence, efficient IDD algorithms and architectures are crucial for enabling high-performance, low-cost MIMO systems.

Expectation propagation (EP) [1] can achieve near-optimal performance for massive MIMO detection. For IDD tasks, the soft iterative EP detector has a natural advantage when integrated with soft decoders, obtaining promising error-rate performance enhancements through turbo IDD iterations. The block-EP (BEP) [2] and double-EP (DEP) [3] IDD schemes facilitate soft message exchange between the EP detector and the decoders, confirming the superiority of EP-based IDD compared to other IDD algorithms. Furthermore, EP-based parallel IDD [4] and EP approximation (EPA) IDD [5] are proposed to reduce latency and alleviate receiver complexity, respectively, at the algorithmic level.

Various hardware architectures for cascaded receivers have been proposed for coded MIMO [6], [7], [8], [9]. In [6], [7], and [8],

IDD architectures integrate minimum mean-square error with parallel interference cancellation (MMSE-PIC) detectors and decoders into soft-input soft-output (SISO) iterative receivers for overall optimization at the cost of high complexity. Conversely, the low-complexity message passing (MP) based receiver [10] with separate detection and decoding (SDD) suffers significant performance degradation.

However, there is a lack of research on the implementation of EP-based IDD architecture to practically demonstrate its superiority. EP-based IDD enhances performance by iteratively updating the approximated posterior distribution in detector with the decoder feedback. Yet, hardware implementations of EP-based detectors [11], [12] simplify message updates considerably, limiting their ability to leverage performance gains from the decoder.

In this brief, we propose a simplified EP approximation-based IDD (sEPA-IDD) receiver for coded MIMO systems. To explore efficient iterative message exchange, a symbol-based simplification for message update is proposed to reduce computational complexity. Moreover, a unified extrinsic message unit (EMU) is designed to enhance hardware utilization. Finally, the first EP-based IDD architecture is implemented with deeply pipelined for high throughput. Implemented with 65 nm CMOS, the proposed IDD receiver attains maximum 0.5 dB gain and achieves 4.03× higher throughput and 6.04× greater area efficiency compared to state-of-the-art (SOA) IDD receivers.

II. PRELIMINARIES

A. System Model

Consider an uplink coded MIMO system with N_t transmitting and N_r receiving antennas. At the transmitter end, each information bit sequence $\mathbf{a}$ is encoded into a codeword. This codeword is then interleaved and modulated into a complex-valued symbol vector $\tilde{\mathbf{x}} \in \tilde{\Theta}^{N_t \times 1}$, where $\tilde{\Theta}$ denotes the constellation set. $\tilde{\mathbf{x}}$ is transmitted through an i.i.d Rayleigh fading channel represented by $\tilde{\mathbf{H}} \in \mathbb{C}^{N_r \times N_t}$, and the received symbol vector is denoted as $\tilde{\mathbf{y}} \in \mathbb{C}^{N_r \times 1}$. By real value decomposition, the system model can be expressed as

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n} \quad (1)$$

where $\mathbf{H} \in \mathbb{R}^{2N_r \times 2N_t}$, $\mathbf{y} \in \mathbb{R}^{2N_r \times 1}$, $\mathbf{n} \in \mathbb{R}^{2N_r \times 1} \sim \mathcal{N}(0, \sigma_n^2 \mathbf{I}_{2N_r})$ is the additive Gaussian white noise, and $\mathbf{x} \in \Theta^{2N_t \times 1}$, where $\Theta = \{\Omega_1, \dots, \Omega_M\}$ is the real-valued constellation set of Q -QAM with $M = \sqrt{Q}$.

B. EP-Based Detectors

The posterior distribution of $\mathbf{x}$ given $\mathbf{y}$ can be expressed as

$$p(\mathbf{x}|\mathbf{y}) \propto \mathcal{N}(\mathbf{y} : \mathbf{H}\mathbf{x}, \sigma_n^2 \mathbf{I}_{2N_r}) \prod_{i=1}^{2N_t} p_D(x_i) \quad (2)$$

where $p_D(x_i) = \sum_{\Omega_j \in \Theta} p_i(\Omega_j) \delta(x_i - \Omega_j)$ is the discrete prior of x_i over Θ . EP detection approximates each $p_D(x_i)$ by a continuous Gaussian term $\hat{p}_D(x_i) = \exp(-(1/2)\Lambda_i x_i^2 + \gamma_i x_i)$ to construct a tractable posterior

$$q(\mathbf{x}) = \mathcal{N}(\mathbf{y} : \mathbf{H}\mathbf{x}, \sigma_n^2 \mathbf{I}_{2N_r}) \exp\left(-\frac{1}{2}\mathbf{x}^T \mathbf{\Lambda} \mathbf{x} + \boldsymbol{\gamma} \mathbf{x}\right) \quad (3)$$

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where $\mathbf{\Lambda} = \text{diag}([\Lambda_1, \Lambda_2, \dots, \Lambda_{2N_t}])$, $\boldsymbol{\gamma} = [\gamma_1, \gamma_2, \dots, \gamma_{2N_t}]$. The mean $\boldsymbol{\mu}$ and variance $\boldsymbol{\Sigma}$ of (3) can be computed as

$$\boldsymbol{\Sigma} = (\mathbf{A} + \boldsymbol{\Lambda})^{-1}, \quad \boldsymbol{\mu} = \boldsymbol{\Sigma} (\tilde{\mathbf{b}} + \boldsymbol{\gamma}) \quad (4)$$

where $\mathbf{A} = \sigma_n^{-2} \mathbf{H}^T \mathbf{H}$, and $\tilde{\mathbf{b}} = \sigma_n^{-2} \mathbf{H}^T \mathbf{y}$.

At sth EP iteration with prior parameter pairs $(\Lambda_i^{[s]}, \gamma_i^{[s]})$, the mean $\mu_i^{[s]}$ and variance $\Sigma_{ii}^{[s]}$ of i th marginal $q^{[s]}(x_i)$ are first calculated by (4). The cavity distribution is computed as $q_E^{[s]}(x_i) = q^{[s]}(x_i)/\hat{p}_D(x_i) \propto \mathcal{N}(t_i^{[s]}, h_i^{2[s]})$. Then, a moment matching (MM) [1] is carried out between $\hat{p}^{[s]}(x_i) = q_E^{[s]}(x_i)p_D(x_i)$ and $q_E^{[s]}(x_i)\exp(-(1/2)\Lambda_i^{[s+1]}x_i^2 + \gamma_i^{[s+1]}x_i)$ to update $(\Lambda_i^{[s+1]}, \gamma_i^{[s+1]})$. Specifically, since $p_D(x_i)$ is discrete, the moments of $\hat{p}^{[s]}(x_i)$ should be obtained by

$$\eta_i^{[s]} = \mathbb{E}_{\hat{p}^{[s]}}[x_i] = a \sum_{\Omega_j \in \Theta} \Omega_j p_i(\Omega_j) q_E^{[s]}(x_i = \Omega_j) \quad (5a)$$

$$\chi_i^{[s]} = \mathbb{V}\text{ar}_{\hat{p}^{[s]}}[x_i] = \mathbb{E}_{\hat{p}^{[s]}}[x_i^2] - (\eta_i^{[s]})^2 \quad (5b)$$

where a is a normalization factor. This involves $2N_t M$ exponentiations to sample over $q_E^{[s]}(x_i) \propto \exp(-((x_i - t_i^{[s]})^2)/(2h_i^{2[s]}))$.

To alleviate the complexity of matrix inversions in (4), EPA [13] reformulates the MM procedure and fixes $h_i^2 = 1/\Lambda_{ii}$ by channel hardening. This approach requires only one inversion for initialization. Each EPA iteration then involves only a discrete distribution inference similar to (5a) and some linear operations, as shown in Algorithm 1 (Lines 2–10).

C. EPA-IDD Receiver

The EPA-IDD receiver executes turbo loops between detector and decoder as described in [5]. Specifically, on the detection side, the priors $p_D(x_i)$ are replaced by the updated extrinsic messages from decoder in the inner loops. In the outer loops, MM assisted by Gaussian projection (GP) is performed to renew the prior parameters $(\mathbf{\Lambda}, \boldsymbol{\gamma})$ based on the decoder outputs, which then restart the inner loops. At the t th outer loop, after S inner loops, the detector produces the extrinsic distribution $q_E^{[t,S+1]}(x_i)$ while $p_D^{[t,1]}(x_i)$ is fed back from the decoder. $(\Lambda_i^{[t+1,1]}, \gamma_i^{[t+1,1]})$ are first updated by MM (Line 17 and 18 in Algorithm 1). If $\Lambda_i^{[t+1,1]} < 0$, GP (Line 20 in Algorithm 1) will be executed instead with the following moments of $p_D^{[t+1,1]}(x_i)$:

$$\vartheta_i^{[t+1,1]} = \mathbb{E}_{p_D^{[t+1,1]}}[x_i], \quad \zeta_i^{[t+1,1]} = \mathbb{V}\text{ar}_{p_D^{[t+1,1]}}[x_i], \quad x_i \in \Theta. \quad (6)$$

The obtained pairs $(\Lambda_i^{[t+1,1]}, \gamma_i^{[t+1,1]})$ are taken as the prior parameters to initialize the $(t+1)$ th outer loop.

In EPA IDD, the quality of the soft messages exchanged between the detector and decoder significantly impacts performance. Although EPA reduces the complexity of matrix inversion, the mean/variance estimations within the exhaustive constellation set for MM and GP still introduce unbearable complexity in massive MIMO systems with high-order modulation. This complexity hinders the efficient implementation of message exchange between detection and decoding.

III. PROPOSED SIMPLIFIED EPA-IDD SCHEME

Inspired by the excellent performance of EP-based IDD, we introduce the first EP-based IDD receiver designed for practical implementation. In this section, we present the sEPA-IDD algorithm,

Algorithm 1 sEPA-IDD Scheme

Input : $\mathbf{A}, \tilde{\mathbf{b}}, \mathbf{v} = \text{diag}(\mathbf{A}), \Lambda_i^{[1,1]}, \gamma_i^{[1,1]}, \beta, \varphi, p_D^{[1,1]}(x_i) = \sum_{\Omega_j \in \Theta} \frac{1}{|\Theta|} \delta(x_i - \Omega_j), \mathcal{M} = \{N_{M1}, N_{M2}, N_G\}$.

Output : $\mathbf{a}$.

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1 for  $t = 1, 2, \dots, T$  do
2   Compute  $\boldsymbol{\mu}^{[t,1]}, \boldsymbol{\Sigma}^{[t,1]}$  using Eq. (4) or wNSA;
3   for  $i = 1, 2, \dots, 2N_t$  do
4      $h_i^{2[t,1]} = \frac{\Sigma_{ii}^{[t,1]}}{1 - \Sigma_{ii}^{[t,1]}\Lambda_i^{[t,1]}}$ ;  $t_i^{[t,1]} = h_i^{2[t,1]}(\frac{\mu_i^{[t,1]}}{\Sigma_{ii}^{[t,1]}} - \gamma_i^{[t,1]})$ ;
5   for  $s = 1, \dots, S$  do
6     for  $i = 1, 2, \dots, 2N_t$  do
7        $\mathcal{D}_{M1} = \text{Sel}_{N_{M1}}(\Theta, q_E^{[t,s]})$ ;
8       Compute  $\eta_i^{[t,s]}$  as Eq. (5) confined to  $\mathcal{D}_{M1}$ ;
9        $\mathbf{t}^{[t,s+1]} = (1 - \beta)\mathbf{t}^{[t,s]} + \beta((\tilde{\mathbf{b}} - \mathbf{A}\boldsymbol{\eta}^{[t,s]})/\mathbf{v} + \boldsymbol{\eta}^{[t,s]})$ ;
10       $\mathbf{h}^{2[t,s+1]} = 1/\mathbf{v}$ ;
11   Compute extrinsic LLRs with  $(\mathbf{t}^{[t,S+1]}, \mathbf{h}^{2[t,S+1]})$ ;
12   Run the BP decoder to output  $p_D^{[t+1,1]}(x_i)$ ;
13    $\mathcal{D}_{M2} = \text{Sel}_{N_{M2}}(\Theta, q_E^{[t,S+1]})$ ;  $\mathcal{D}_G = \text{Sel}_{N_G}(\Theta, p_D^{[t+1,1]})$ ;
14   Compute  $\eta_i^{[t+1,1]}, \chi_i^{[t+1,1]}$  as Eq. (5) confined to  $\mathcal{D}_{M2}$ .
15   Compute  $\vartheta_i^{[t+1,1]}, \zeta_i^{[t+1,1]}$  as Eq. (6) confined to  $\mathcal{D}_G$ .
16   if  $\chi_i^{[t+1,1]} < h_i^{2[t,S+1]}$  then
17      $\Lambda_i^{[t+1,1]} = \beta(\frac{1}{\chi_i^{[t+1,1]}} - \frac{1}{h_i^{2[t,S+1]}}) + (1 - \beta)\Lambda_i^{[t,1]}$ ;
18      $\gamma_i^{[t+1,1]} = \beta(\frac{\eta_i^{[t+1,1]}}{\chi_i^{[t+1,1]}} - \frac{t_i^{[t,S+1]}}{h_i^{2[t,S+1]}}) + (1 - \beta)\gamma_i^{[t,1]}$ ;
19   else
20      $\Lambda_i^{[t+1,1]} = \varphi \frac{1}{\zeta_i^{[t+1,1]}}$ ;  $\gamma_i^{[t+1,1]} = \varphi \frac{\vartheta_i^{[t+1,1]}}{\zeta_i^{[t+1,1]}}$ ;

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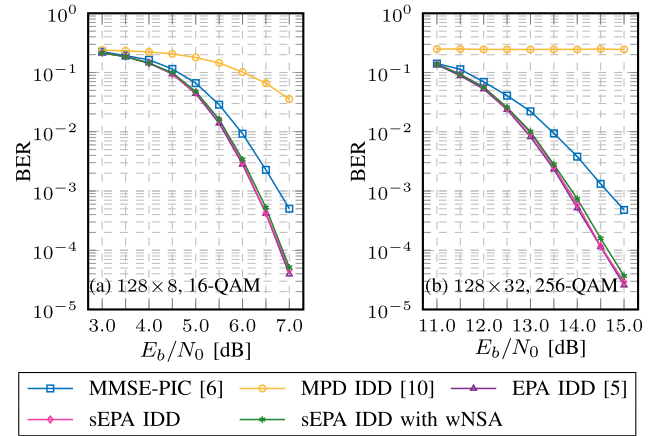


Fig. 1. BER performance of sEPA IDD and sEPA IDD with wNSA, compared with MMSE-PIC IDD, MPD IDD, and EPA IDD. The soft-in design in MPD IDD refers to [6].

which reduces the complexity of MM and GP in EPA-IDD to make it more implementation-friendly while maintaining high performance.

In EPA-IDD, (5) and (6) require numerous exponentiations and multiplications. To address this, we propose a constellation point selection method to construct a configurable set $\mathcal{D} \subseteq \Theta$ for simplified mean and variance estimations

$$\mathcal{D} = \text{Sel}_N(\Theta, p) := \arg \max_{\Omega_i \in \Theta, |\mathcal{D}|=N} \sum_{\mathcal{D}} p(\Omega_i) \quad (7)$$

where p is a distribution over Θ . In this way, $\mathcal{D}$ retains the N most reliable constellation points measured by p . The moments estimations in GP and MM can then be confined to the subset $\mathcal{D}$ instead of the

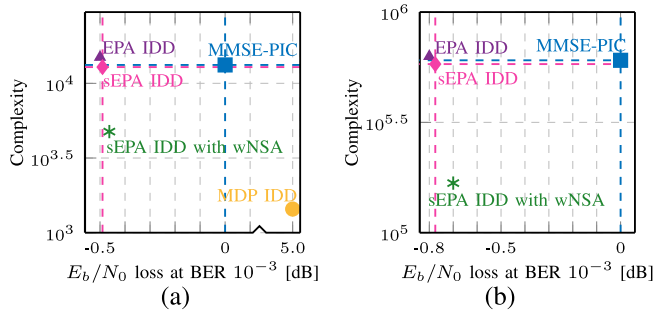


Fig. 2. Comparison of the trade-off between complexity and performance loss at $\text{BER} = 10^{-3}$. The multiplications and exponentiations are considered, and MMSE-PIC is set as the performance benchmark. (a) 128×8 , 16-QAM. (b) 128×32 , 256-QAM.

whole constellation Θ , which enable efficient symbol/bit interface designs.

A. Configurable Set for MM With Gaussian Distribution

For MM, we can confine the computation of (5) to a configurable constellation set $\mathcal{D}_M = \text{Sel}_{N_M}(\Theta, q_E)$. Since q_E is Gaussian, the constellation points closer to the mean of $q_E(x_i)$, t_i , should be considered more reliable. $\mathcal{D}_M$ should then be composed of the N_M nearest constellation points to t_i . Thus, constructing $\mathcal{D}_M$ requires up to M comparisons.

B. Configurable Set for GP With Bit-Wise Decoder Feedback

We also confine the computation of GP (6) to a set $\mathcal{D}_G = \text{Sel}_{N_G}(\Theta, p_D)$, which is constructed using the bit-wise log-likelihood ratio (LLR) feedback from the decoder. Each $\Omega \in \Theta$ is labeled by a sequence of $\log_2 M$ bits under Q -QAM, denoted by $b(\Omega)$. For each x_i , we obtain a bit sequence $b(\Omega^0)$ corresponding to $\Omega^0 \in \Theta$ by making a hard decision on the corresponding LLRs; Ω^0 is considered the most reliable symbol. Symbols with bit sequences closer to $b(\Omega^0)$ should be more reliable. Thus, we define a series of subsets of Θ as $\mathcal{U}_k := \{\Omega \in \Theta | d_H(b(\Omega), b(\Omega^0)) = k\}$, $k = 0, \dots, \log_2 M$, where $d_H(\cdot)$ denotes the Hamming distance. Each $\mathcal{U}_k$ contains $\binom{\log_2 M}{k}$ symbols. The set $\mathcal{D}_G$ is then constructed by

$$\mathcal{D}_G = \bigcup_{k=0}^l \mathcal{U}_k \cup \text{Sel}_{N_G - \sum_{k=0}^l \binom{\log_2 M}{k}} \left(\binom{\log_2 M}{k} \right) (\mathcal{U}_{l+1}, p_D) \quad (8)$$

where $l = \arg \max_l \{\sum_{k=0}^l \binom{\log_2 M}{k} \leq N_G\}$. This bit-wise construction reduces the comparison complexity from $\mathcal{O}(M \log_2 M)$ to $\mathcal{O}(\binom{\log_2 M}{l+1} \log_2 \binom{\log_2 M}{l+1})$.

The proposed sEPA IDD algorithm is outlined in Algorithm 1. In the inner loop, $\mathcal{D}_{M1}$ for MM selects N_{M1} points. In the outer loop, $\mathcal{D}_{M2}$ for MM and $\mathcal{D}_G$ for GP select N_{M2} and N_G points, respectively. We summarize them by a set $\mathcal{M} = \{N_{M1}, N_{M2}, N_G\}$. Compared to the original EPA IDD, sEPA IDD reduces computations from $2N_t MT(S+1)$ exponentiations and $2N_t MT(2S+7)$ multiplications to $2N_t T(N_{M1}S + N_{M2})$ exponentiations and $2N_t T(2N_{M1}S + 4N_{M2} + 3N_G)$ multiplications in constellation point processing. To further reduce complexity, we implement matrix inversion using the weighted Neumann series approximation (wNSA) [11], which involves $\mathcal{O}(N_t^2)$ complexity.

IV. SIMULATION RESULTS

In this section, simulation results and analysis are presented to verify the efficiency of sEPA IDD. We consider a length-1024 rate-1/2 polar code and a nonreset BP decoder with decoding iterations

TABLE I
UNIFIED FUNCTION FOR $f(t_i)$ AND $f_e(t_i)$

	Unified Function	$f(t_i)$		$f_e(t_i)$	
		k	Range	Ω_{e0}	Range
BPSK	$2t_i$	0	all t_i	-1	all t_i
QPSK	$2t_i$	0	all t_i	-1	all t_i
16-QAM	$2t_i$	0	$ t_i \leq 2$	-1	$ t_i \leq 1$
	$4t_i - 4\text{sgn}(t_i)$		$ t_i > 2$	-	-
	$4 - 2 t_i $	1	all t_i	-3 or 3	$ t_i > 1$
64-QAM	$2t_i$	0	$ t_i \leq 2$	-1	$ t_i \leq 1$
	$4t_i - 4\text{sgn}(t_i)$		$2 < t_i \leq 4$	-	-
	$6t_i - 12\text{sgn}(t_i)$		$4 < t_i \leq 6$	-	-
	$8t_i - 24\text{sgn}(t_i)$		$ t_i > 6$	-	-
	$-4 + 2 t_i $	1	$ t_i \leq 4$	-1 or 1	$1 < t_i \leq 3$
	$12 - 2 t_i $		$ t_i > 4$	-7 or 7	$ t_i > 5$
	$12 - 4 t_i $	2	$ t_i \leq 2$	-	-
	$8 - 2 t_i $		$2 < t_i \leq 6$	-5 or 5	$3 < t_i \leq 5$
	$20 - 4 t_i $		$ t_i > 6$	-	-

$D = 2$. The numbers of outer and inner loops for all IDD algorithms are set to $T = 3$ and $S = 1$, respectively. Fig. 1 compares the bit error ratio (BER) performance of sEPA IDD with MMSE-PIC IDD, MPD IDD, and EPA IDD. The BER performance of sEPA IDD with k -term wNSA is also investigated. Fig. 2 shows the trade-off between complexity and performance of the results in Fig. 1. For $N_r \times N_t = 128 \times 8$, 16-QAM, sEPA IDD with $\mathcal{M} = \{1, 2, 2\}$ can achieve performance close to EPA IDD with 86% complexity. Additionally, by introducing three-term wNSA, sEPA IDD maintains similar BER performance while saving 68% complexity of EPA IDD and meanwhile outperforms MMSE-PIC IDD for 0.5 dB at $\text{BER} = 10^{-3}$. For $N_r \times N_t = 128 \times 32$, 256-QAM, sEPA IDD with $\mathcal{M} = \{2, 2, 2\}$ shows near-EPA-IDD performance. With 12-term wNSA, sEPA IDD shows around 0.1 dB loss compared to EPA IDD with 27% complexity of EPA IDD but outperforms MMSE-PIC IDD for 0.7 dB at $\text{BER} = 10^{-3}$. Moreover, sEPA IDD significantly outperforms MPD IDD for all configurations.

V. HARDWARE IMPLEMENTATION

In this section, an efficient hardware architecture is designed for the sEPA-IDD algorithm which supports a 128×8 polar-coded MIMO system with 16-QAM modulation. The code length is 1024 and the code rate is 1/2. The numbers of iterations are set as $S = 1$, $D = 2$, $T = 3$, and $\mathcal{M} = \{1, 2, 2\}$.

A. System Architecture

Fig. 3(a) depicts the top-level architecture of the sEPA-IDD receiver, which is divided into four parts: an EPA detector, a BP decoder, an interface unit, and a joint processing unit. By reasonable computation parallelism, the dataset is partitioned into 32 sets, except for the BP decoder. The interface consists of a unified EMU, a bit-wise prior distribution unit (PDU), a serial-in-parallel-out (SIPO) register, and a parallel-in-serial-out (PISO) register, which achieves efficient message exchange. To support hardware sharing, the extrinsic symbol distributions are condensed into $\mathbf{t}$, and will be decompressed through EMU after decoding.

B. Unified EMU

Incorporated with the efficient memory access, the unified EMU is responsible for extrinsic message from the detector with homologous input $\mathbf{t}$.

- 1) Computation of extrinsic bit LLRs which are delivered to the decoder through SIPO register as the red line shown in Fig. 3(a).

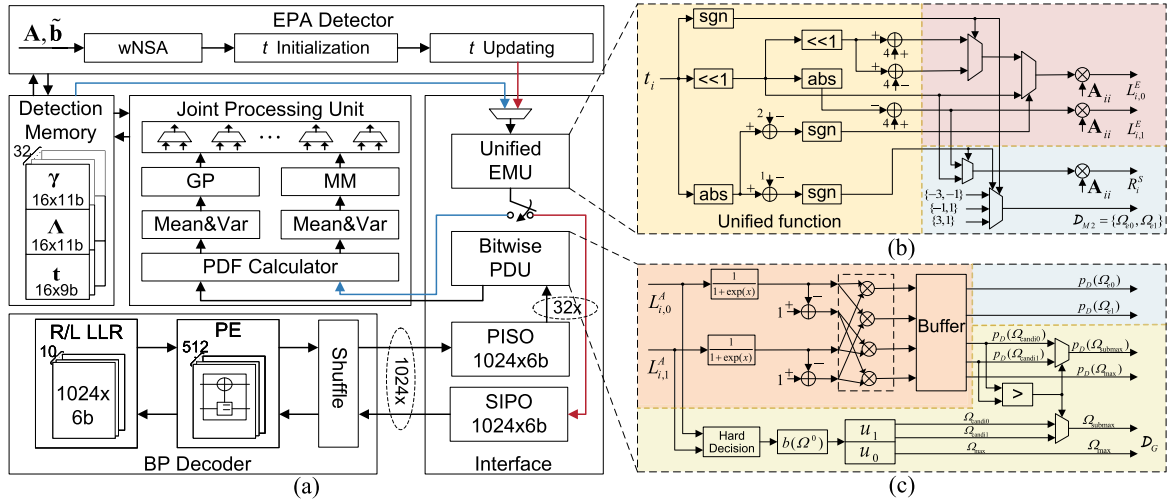


Fig. 3. Architecture of the sEPA-IDD receiver. (a) Top-level architecture. (b) Architecture of unified EMU. (c) Architecture of bit-wise PDU.

TABLE II
COMPARISON WITH SOA JOINT DETECTION-DECODING RECEIVER DESIGNS

Receiver Type	This work IDD		Sun [7] IDD		Tang [8] IDD		Chen [10] SDD	
Algorithm	EPA Detector	Polar Decoder	MMSE-PIC Detector	LDPC Decoder	MMSE-PIC Detector	NBLDPC Decoder	MP Detector	Polar Decoder
Antennas ($N_r \times N_t$)	128 × 8	-	4 × 4	-	4 × 4	-	128 × 8	-
Modulation	16-QAM	-	256-QAM	-	256-QAM	-	QPSK	-
Block Length (bits)	-	1024	-	1944	-	416	-	1024
Technology (nm) @ (Voltage)	65 (@1.0 V)		40 (@0.9 V)		65 (@1.0 V)		40 (@0.9 V)	
Frequency (MHz)	675		333 (204.9 [†])		517		307	
Area (mm ²)	3.20		1.97 (5.20 [†])		0.7* (2.8 [‡])		0.076* (0.20 [‡])	
Norm. Throughput (Gb/s)	3.07		0.76 [†]		1.84 [‡]		4.92 [‡]	
Norm. Energy (pJ/b)	169.6		319.5 [†]		57.7 [‡]		31.7 [‡]	
Norm. TAR (Gb/s/mm ²)	0.96		0.15 [†]		0.66 [‡]		24.53 [‡]	

* Memory for information exchange excluded.

[†] Scaled to 65 nm and 1.0 V CMOS process with frequency $\propto s$, area $\propto 1/s^2$ and power $\propto (1/s)(1.0/V_{dd})^2$, where $s = \text{Technology}/65 \text{ nm}$.

[‡] Normalized to 128 × 8 configuration: throughput $\propto (8/N_t) \times (\log_2 N_t / \log_2 8)$, and power and area $\propto (8/N_t)^2$, while scaled to 65 nm Technology and 1.0 V.

2) Decompression of extrinsic symbol distributions, which are passed to the joint processing unit as the blue line shown in Fig. 3(a).

By max-log algorithm, the bit LLR of k th bit for x_i , $L_{i,k}^E$, can be approximated as $L_{i,k}^E = \mathbf{A}_{ii} f(t_i)$, with

$$f(t_i) = \frac{1}{2} \left(\min_{\Omega \in \mathcal{A}_k^0} |t_i - \Omega|^2 - \min_{\Omega \in \mathcal{A}_k^1} |t_i - \Omega|^2 \right) \quad (9)$$

where $\mathcal{A}_k^b$ denotes the subset of Θ whose k th bit is b . $f(t_i)$ can be obtained by a piecewise linear function [14]. On the other hand, since $N_{M2} = 2$, $\mathcal{D}_{M2} = \{\Omega_{e0}, \Omega_{e1}\}$ with $R_i^S = \ln(q_E(\Omega_{e1})/q_E(\Omega_{e0})) = \mathbf{A}_{ii} f_e(t_i)$, where

$$f_e(t_i) = \frac{1}{2} \left(|t_i - \Omega_{e0}|^2 - |t_i - \Omega_{e1}|^2 \right). \quad (10)$$

Then, $q_E(\Omega_{e0})$ and $q_E(\Omega_{e1})$ can be decompressed from R_i^S .

It is worth noticing that $f_e(t_i)$ and $f(t_i)$ have same structure, enabling the design of a unified calculator. The essence of $f(t_i)$ is to find the nearest points to t_i in $\mathcal{A}_k^b$ and $\mathcal{A}_k^1$, which implicitly includes the cases of $(\Omega_{e0}, \Omega_{e1})$ selected to calculate $f_e(t_i)$. Therefore, we derive a unified piecewise linear function for $f(t_i)$ and $f_e(t_i)$ as shown in Table I. As illustrated in Fig. 3(b), the unified EMU

is partitioned into three parts, including the generation of unified functions, the output for extrinsic bit LLRs, and the output for extrinsic symbol distributions. With the same generation part, the outputs for both modes can be determined by selectors, without additional hardware cost.

C. Bit-Wise PDU

Fig. 3(c) shows the block diagram of the bit-wise PDU that calculates $p_D(x_i)$ and constructs the configurable set $\mathcal{D}_G$. The probabilities of four symbols are computed by multipliers and stored in the buffer. Meanwhile, $b(\Omega^0)$, $\mathcal{U}_0$, and $\mathcal{U}_1$ can be generated. After comparing the probabilities of two candidates in $\mathcal{U}_1$, $\mathcal{D}_G$ is obtained. Finally, the outputs provide the required $p_D(x_i)$ for MM and GP, respectively.

VI. IMPLEMENTATION RESULTS AND COMPARISON

To demonstrate the efficiency of the sEPA-IDD receiver, it is synthesized under TSMC 65 nm CMOS process by Synopsys Design Compiler. Fig. 4 shows the floating-point (Fp) and quantization (Qtz) performance of the proposed receiver. The bit lengths of main parameters are shown in Fig. 3(a). Fig. 5 presents the timing schedule of the sEPA-IDD receiver architecture. The processing latency of each module and the serial output latency of the 31 datasets are distinguish

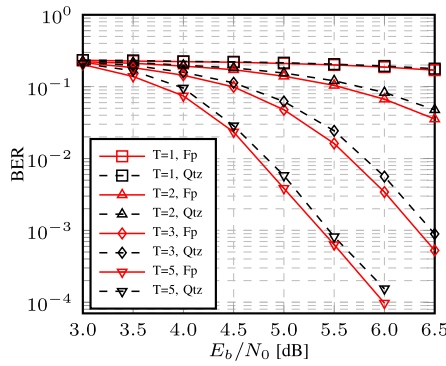


Fig. 4. Quantized BER performance of the sEPA-IDD receiver.

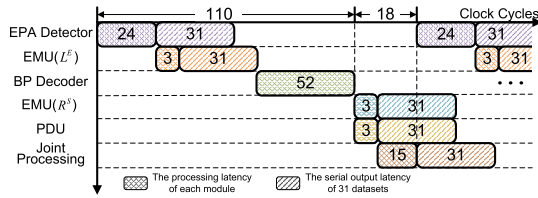


Fig. 5. Timing schedule of the proposed sEPA-IDD receiver.

by shading. The dataflow between the decoder and other modules is aligned by SIPO and PISO registers. The total latency for T outer loops is $110T + 18 \times (T - 1)$. To improve the throughput and energy efficiency, an early stopping criterion based on $\mathbf{G}$ matrix [15] is adopted. The overall receiver occupies a core area of 3.20 mm^2 and supports a throughput of 3.07 Gb/s at clock frequency 675 MHz with an energy-efficiency of 169.6 pJ/b when $\text{BER} = 10^{-3}$.

The comparisons of the proposed IDD receiver and the existing receivers are concluded in Table II. Since our receiver focus on the integration of detector and decoder to achieve error-correction performance gains, emphasizing deep pipelining and interface optimizations, only receivers with cascaded structure are considered. While [10] demonstrates higher area efficiency, it is worth noting that it adopts QPSK modulation and lacks memory allocation for information exchange, which only supports SDD. Our sEPA-IDD receiver achieves a $4.03\times$ higher throughput and $6.40\times$ higher area efficiency than those of [7]. Compared with [8], the proposed receiver can achieve $3.00\times$ higher throughput and $1.60\times$ higher area efficiency.

VII. CONCLUSION

In this brief, we propose a sEPA-IDD receiver for polar-coded MIMO systems, which has lower computation complexity than the original EPA-IDD receiver by a symbol-based simplification. Moreover, a deeply pipelined hardware implementation on 65 nm CMOS for 128×8 , 16-QAM massive MIMO exhibits higher efficiency

than existing IDD receivers. This proposed IDD scheme can also be applied to LDPC-coded MIMO systems. Future work will be directed toward optimizing the scheduling of detector and decoder to reduce latency.

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